

Interjections and Interactions in the German Parliament: An Exploratory Study Using NLP to Analyze Parliamentary Dynamics

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Abstract

This project examines interactions in the German parliament as represented by verbal and nonverbal interjections. It provides a publicly accessible tool¹ to process XML files and extract relevant information from parliamentary protocols, including speaker and interjector names, party affiliation, gender, and the type of interjection. Based on this information, we conduct exploratory studies; for instance, we find no significant differences in the use of interjections by and towards male and female MPs. Furthermore, we use a fine-tuned BERT model by Reinig et al. (2024) to demonstrate how machine learning can be applied to analyze large datasets qualitatively. Lastly, we investigate how parties mention each other and (unsurprisingly) find that they portray themselves in a positive light (eliciting supportive interjections from their own ranks) while representing political opponents more negatively, who, in turn, defend themselves through interjections. This project lays the groundwork for exploring many more interesting research ideas.

1 Introduction

Despite their potential negative reputations, interjections are a vital aspect of parliamentary speech. As Burkhardt (1993) notes for Germany, communication in parliament can only be regarded as spoken speech to a certain extent, as most speakers rely on pre-written texts for their presentations. For this reason, it is particularly compelling to examine actual "spontaneous" contributions as well as possible reactions to those contributions that are not as staged: interjections². Interjections give those without the right to speak an opportunity to

react to what they hear, thereby adding a dialogical element to the often-monologous parliamentary speech (Burkhardt, 1993).

The significance of interjections as a parliamentary instrument for communication is evident in the fact that, while the rules of procedure do not explicitly allow them, they persist as an old parliamentary custom. Interjections are permitted and even stenographed (Deutscher Bundestag). This means that, unlike official questions during a speech, interjections do not require the Bundestag president's prior approval. However, hecklers can also face admonishments for their interruptions, with calls to order by the president potentially leading to the withdrawal of the floor or a fine of up to €2,000 in repeated cases (Bundesministerium der Justiz; Deutscher Bundestag, 2021).

So why even take the risk? Burkhardt (1993) highlights two aspects of interjections. On the one hand, he notes that growing media coverage and the pressure to acquire renown have led to a debate style in parliament that prioritizes disagreement with political opponents and alignment with public opinion over cross-party cooperation. Verbal interjections in parliament can therefore be considered multi-addressed speech acts (*mehrfachadressierte Sprachhandlung*, a term coined in German by Kühn (1983)), reaching not only parliamentarians but also, crucially, the broader audience of voters consuming media coverage (e.g., through newspapers, radio, television, or, what Burkhardt could not have known in 1993, on social media). Consequently, actual decision making processes have shifted to committees whose discussions and records are generally inaccessible to the public. Burkhardt (1993) thus questions whether verbal interjections are not to be evaluated positively, since they are one of the few surviving true dialogical elements of political debate.

On the other hand, he criticizes how, interjections have become less about challenging and/or

¹The GitHub repository is publicly available at <https://github.com/hannahsteinbach/Interjections-German-Parliament>

²Burkhardt (1993) uses the term 'Zwischensignale,' which can be translated as 'interjections.' In this text, we will refer to Zwischenruf as 'verbal interjection' and other interruptions, such as applause, as 'nonverbal interjections.'

expressing disagreement or agreement with the speaker and more about flustering them or publicly discrediting their position. He also mentions how their quantity and nature depends on the subject of the debate and the degree of disagreement between the parties. Thus, interjections are also used strategically to expose the speaker to the public.

This perspective aligns with [Hitzler \(1990\)](#), who emphasizes that parliamentary discourse is not aimed at persuading opponents but rather at stigmatizing them and gaining voter support. [Hitzler \(1990\)](#) compares political discourse to a game, where power can be won by those who excel as performers—be it through sharp thinking, quick reactions, or stealing the spotlight from the speaker. In this sense, politicians resemble actors whose ability to "win" depends on their performative skills.

The goal of this paper is to explore this "game of power" and analyze interactions in the German Bundestag. Specifically, we aim to assess whether the dynamics described by [Burkhardt \(1993\)](#) and [Hitzler \(1990\)](#) have changed or been reinforced now, 30 years after. This analysis will encompass several aspects of parliamentary communication, including—but not limited to—interjections. Drawing on relevant literature, we will also consider gendered discourse and the potential for leveraging technologies, such as large language models, to facilitate the analysis of vast amounts of parliamentary data. This paper further seeks to illustrate the richness of parliamentary debates as a research subject, highlighting the many angles of analysis they offer. Accordingly, this work is not intended to be comprehensive but rather a glimpse into the possibilities.

The contributions of this paper are as follows:

- A ready-to-use pipeline for processing official German parliamentary documents into data frames, enabling the extraction of a wide range of information, such as speaker gender and the presence of verbal and nonverbal interjections, as well as interjector gender (see [Appendix A](#) for the full list).
- Contextualized analyses of interjections and interactions in parliament, informed by relevant literature.
- An evaluation of machine learning methods for automating the annotation of political speeches.

- Manually annotated data (utterance speech acts, interjection types)

2 Related Work

Parliamentary speech has been analyzed in various ways across different times and countries. Numerous tools exist to help processing official documents. For the German parliament, the [PolMine](#) project aims to transform available text into easily accessible, digitized corpora. Within this project, the [GermaParl \(Blätte, 2017\)](#) was developed based on the protocols of plenary debates that are published by the German Bundestag. They automatically process these protocols into XML files, modeled on the standards of the [Text Encoding Initiative](#). Similar initiatives have been carried out for different countries, such as Sweden ([Yrjänäinen et al., 2024](#)) or the Netherlands, Flanders and Belgium ([Marx and Schuth, 2010](#)). [Hladka et al. \(2020\)](#) provide a comprehensive overview of comparable projects in other countries.

More specifically related to this project, [Wisik \(2021\)](#) explores interruptions, examining how they are recorded in official parliamentary records and discussing whether their encodings in currently available digital corpora are suitable for automated quantitative analysis.

Before the creation of these digital parliamentary corpora, [Burkhardt \(1993\)](#) compared interjections across one session from the Weimar National Assembly of 1919, one session from the 1st Federal Parliament in 1951, and one session from the 11th German Federal Parliament in 1989. Although he notes that his analysis is exemplary and not statistically representative due to the limited material, he also argues that the results align with his general impressions of debates from different time periods. He differentiates between six types of verbal interjections:

- *affirmativa*: expressions of (partial) agreement
- *erotetika*: informational questions
- *direktiva*: demands and requests
- *memoranda*: rhetorical questions or expressions intended to elicit a reaction from the speaker
- *dissentiva*: expressions of disagreement
- *evaluativa*: moral judgments or discrediting of the content or speaker

[Burkhardt \(1990\)](#) concludes that while the frequency, length and variability of interjections as

well as judgmental interjections and reactions increased over time, interjections concerning parliamentary procedure decreased over the years. He specifically observes a shift from affirmativa (positive, supportive verbal interjections) to memoranda, dissentiva, evaluativa (negative, critical, or ridiculing verbal interjections). The fact that a comparison of just three debates yields such interesting results underlines the importance of having simple ways to reliably, objectively, and statistically analyze larger amounts of data.

Once processed appropriately, digital parliamentary corpora enable researchers to address important questions about political discourse and its societal relevance. For example, [Vögele and Thoms \(2019\)](#) build on a project by the *Süddeutsche Zeitung* examining the AfD's interactions during their first six months in the German Bundestag, including their use of verbal and nonverbal interjections ([Brunner et al., 2019](#)). The study finds that both the behavior of the AfD and the responses from the remaining parties contribute to the party's isolation in parliament, as they are not only excluded by others but also actively distance themselves ([Brunner et al., 2019](#)). [Vögele and Thoms \(2019\)](#) extends this analysis to the Baden-Württemberg state parliament, focusing on protocols from the 15th and parts of the 16th election periods. Key findings include that AfD members or non-affiliated MPs that used to be AfD members are virtually the only ones applauding for the party. The AfD also frequently uses laughter as a nonverbal interjection to discredit political opponents.

More generally, they find that negative (criticizing) verbal interjections outweigh positive (supportive) ones. All parliamentary groups mainly use supportive interjections for speakers from their own faction or coalition, while opposition parties predominantly criticize government speakers. In the opposition, CDU is the most active interjecting party. The AfD's interjections primarily distance themselves from government parties (15th period: Greens and SPD; 16th period: CDU and Greens), with the FDP being relatively spared. Although [Vögele and Thoms \(2019\)](#) analyze reactions to interjections and find that 90% are ignored by speakers, [Hitzler \(1990\)](#) notes that the range of possible reactions to verbal interjections is too diverse for all of them to be documented in the protocol, making it rather misleading to interpret whether or not someone reacts to an interjection, as nonverbal aspects

like gestures or facial expressions are probably not fully recorded.

A further interesting finding by [Vögele and Thoms \(2019\)](#) is that politicians with higher positions in parliamentary groups, such as (deputy) chairpersons or parliamentary secretaries, are particularly active in making verbal interjections. This may be part of their role, but the authors also note that these individuals often sit closer to stenographers, making their interjections more likely to be recorded. Lastly, they question whether the AfD's isolated position helps reinforce their self-image as victims while consciously distancing themselves from the "established parties."

Another common topic in research on parliamentary debates is gender. [Miller and Sutherland \(2023\)](#), for instance, examined speech patterns during congressional committee hearings from 1994 to 2018 in the US. They found that women are approximately 10% more likely to be interrupted in Senate committees; in hearings discussing women's issues, they were more than twice as likely to be interrupted than in other hearings.

Similar studies of the German parliament include [Burkhardt \(1990\)](#); [Irmer and Versloot \(2024\)](#); [Ash et al. \(2024\)](#). [Burkhardt \(1990\)](#) found that both in the German National Assembly in 1990 and in the Bundestag in 1983/1984, women were more likely to be interrupted than men. However, as [Irmer and Versloot \(2024\)](#) points out, [Burkhardt \(1990\)](#) based his findings on a random sample that cannot be generalized and did not conduct any statistical tests to support his conclusions. [Irmer and Versloot \(2024\)](#) and [Ash et al. \(2024\)](#) present contrary results.

[Irmer and Versloot \(2024\)](#) analyzed verbal interjections from 2019, finding that most were negative but that female speakers did not receive proportionally more negative interjections than male speakers. [Ash et al. \(2024\)](#) analyzed both verbal and nonverbal interjections in all 16 German state parliaments between 1991 and 2020. They also found that women were not subjected to more interruptions overall. In fact, women received more positive and fewer negative reactions than men. When women spoke on "feminine" topics, they received more applause than men, and vice versa. However, they also noted that women's speeches received fewer reactions overall, arguing that "the ultimate negative reaction is not necessarily a derogatory response from the opposition but rather no reaction at all" ([Ash et al., 2024, p.4](#)).

For the Austrian parliament, [Hofmann and Wisnik \(2024\)](#) examined parliamentary records of the National Chamber from 1996 to 2017. They found that discourse styles varied over time, especially with changes in government, and highlighted that style depended on party affiliation, gender, and utterance type (speech vs. interjection). In terms of gender, they found that female MPs used more positive and arousing language, whereas men were responsible for a higher percentage of verbal interjections.

These findings demonstrate that the analysis of political discourse is highly context-dependent, and results cannot easily be generalized across countries or time periods. This section has highlighted various aspects of parliamentary speech analysis, with a specific focus on interruptions, interjections, and gender as a subtopic. We will now move on to our own experiments.

3 Data and Task

For this project, we examine German parliament protocols from the 19th legislature period (2017–2021), as these are the most recent ones available as XML files provided by GermaParl ([Blätte, 2017](#)). Our dataset comprises 239 sessions in total. We focus exclusively on speeches and contributions made by members of parliament, excluding presidential speeches and speeches by members of the government.

The 19th legislature period is particularly interesting because it marks the first time the AfD entered the Bundestag. Building on earlier studies—[Brunner et al. \(2019\)](#), who analyzed the first six months of this period, and [Vögele and Thoms \(2019\)](#), who studied interactions in the Baden-Württemberg parliament—we aim to investigate whether similar patterns of (self-)isolation of the right-wing populist party can be observed. Another area of interest is gender. After peaking at around 36.3% during the 18th parliamentary term, the proportion of women dropped to approximately 30.9% in the 19th, marking the lowest percentage since the 14th parliamentary term, which began in 1998 ([Bundeszentrale für politische Bildung, 2017](#)). In line with [Irmer and Versloot \(2024\)](#) and [Ash et al. \(2024\)](#), we also examine whether there are significant differences in how female MPs are treated in terms of interjections and interruptions.

To analyze the data from the 19th legislature period, we processed the XML files provided by

[Blätte \(2017\)](#) using a script based on Regular Expressions. Due to the standardized structure of the protocols, this method was sufficient, with manual adjustments required only in exceptional cases.

For each paragraph, we extracted the following information:

- **General Information:** Name of the debate file, legislature period, date, name of the agenda, total number of speeches, paragraph number within the speech.
- **Speaker-Specific Information:** Name of the speaker, party of the speaker, role of the speaker.
- **Classification of the Paragraph:** Whether it is an interjection or part of the “official” speech.
- **Interjection-Specific Information:** Non-verbal or verbal interjection? Specific type: "Beifall", "Lachen", "Heiterkeit", "Unruhe" (nonverbal), "Zwischenruf", "Gegenruf", "Widerspruch", "Zustimmung" (verbal).
- **Interjector-Specific Information (if identified):** Name of the interjecting person, party of the interjecting person.

Since gender is not included in the protocol, we explored alternative methods. [Ash et al. \(2024\)](#), for example, highlighted the challenges of coding gender based solely on names. They used Wikipedia articles to assign gender based on the frequency of female and male pronouns, manually verifying their results. In our study, we leveraged the [Bundestag Stammdaten](#), which contain biographical data for all MPs since 1949. We wrote a script to match the names in the protocol with the names in the Stammdaten to extract gender. Although we have not done so yet, this method could also be used to retrieve other relevant information, such as age, academic titles, profession, religion, marital status, or place of birth. This connection to the Stammdaten thus offers valuable opportunities for further analysis.

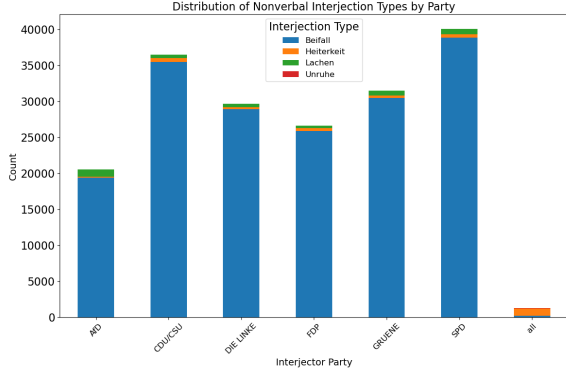


Figure 1: Distribution of Nonverbal Interjection Types

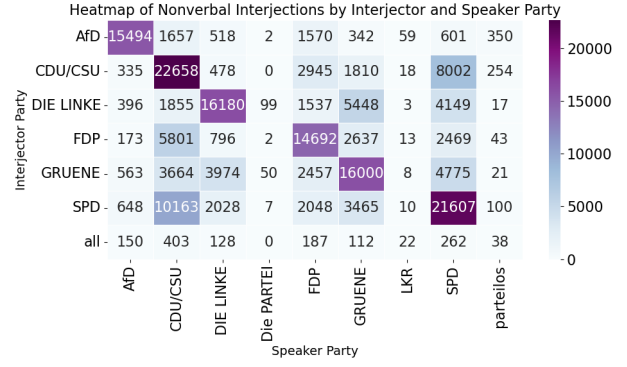


Figure 2: Heatmap of nonverbal interjections

4 Experiments

4.1 Interjection Analyses

4.1.1 Nonverbal Interjections

As seen in Figure 1, the vast majority of nonverbal interjections is applause. This holds true for all parties, with the exception being when MPs from all parties react together, where "Heiterkeit" (amusement) dominates. We also observe that the government parties, CDU and SPD, have the highest number of nonverbal interjections given. However, the differing sizes of the fractions must be considered. To account for this, we normalize the number of nonverbal interjections per party by its number of seats. This same methodology was applied by Vögele and Thoms (2019). A different picture emerges here: Table 1 shows that, when normalized by the number of MPs, the Greens produce the most nonverbal interjections. We see that, in general, opposition parties make more nonverbal interjections than the government parties, with one exception: the AfD, which has approximately half the number of interjections per MP as the Greens. These numbers suggest that, with the exception of the AfD, opposition parties are generally more active in giving nonverbal interjections than government parties.

Party	Count	Num. of MPs	Interj. per MP
GRÜNE	31512	67	470.33
DIE LINKE	29684	69	430.20
FDP	26626	80	332.83
SPD	40076	152	263.66
AfD	20593	88	234.01
CDU/CSU	36500	246	148.37

Table 1: Nonverbal Interjections by Party (19th period)

The heatmap in Figure 2 clearly illustrates how nonverbal interjections are predominantly used

within a party, as the vast majority is applause. Apart from their own party, we also see that the CDU gives a lot of nonverbal interjections (i.e., most probably applause, as we have observed) to the SPD, and vice versa, indicating possible support within the coalition. We also observe significant exchanges between the Greens and the Left Party, the Greens and the SPD, as well as the FDP and CSU/CDU, which may suggest alignment in support based on ideological similarities.

To gain a deeper insight into party interactions, we must also investigate which parties the individual parties react to nonverbally, and from which they receive nonverbal interjections. All plots can be found in the Appendix A.1. By examining all types of received and given interjections per party, we see that MPs from every party primarily receive from and give applause to their own party. However, they still give and receive applause from other parties. For the AfD, in particular, we see that 80% of the applause given by the AfD goes to their own party (Figure 15), which is higher than for any other party. Almost 100% of the applause received by the AfD comes from the AfD itself (Figure 21). Thus, while all other parties exhibit a more balanced distribution of applause given and received, the AfD receives even less (positive) engagement than it gives to other parties. This illustrates how the AfD is isolated by other parties and also isolates itself.

4.1.2 Verbal Interjections

As a first step, we examine the verbal interjections to obtain a general overview. We tokenized all verbal interjections but excluded cases where only the speech act (e.g., "Widerspruch" or "Zuruf") was stenographed in the protocol. Instead, we included only instances where the actual words spoken were

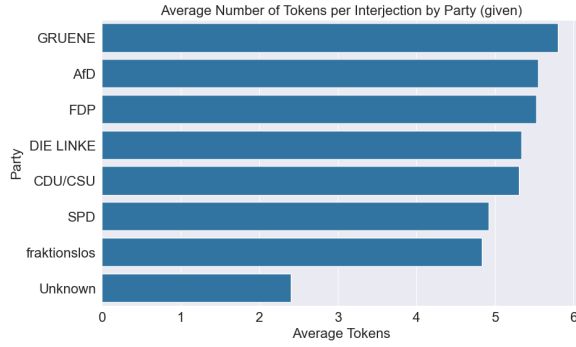


Figure 3: Avg. Number of Tokens per Interjection

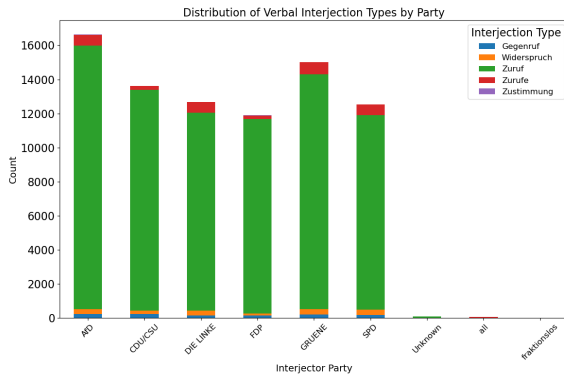


Figure 4: Distribution of Verbal Interjection Types

transcribed. We find that each interjection consists of an average of 5.4 tokens. Comparing the length of interjections by female MPs (5.43 tokens) and male MPs (5.7 tokens), we observe a minimal difference. However, verbal interjections where the interjector could not be identified consist of only 2.98 tokens on average. This is likely because such interjections are too short or indistinct for the stenographers to attribute them to a specific MP. As shown in Table 3, opposition parties (with the Greens leading, followed by the AfD, FDP, and the Left Party) generally produce longer interjections than government parties (CDU/CSU, SPD), and non-affiliated MPs. The difference is however small. As observed earlier, MPs whose party affiliation is unknown since they are unknown tend to have shorter interjection.

Looking at the distribution of interjection types in Figure 4, we see that the AfD is responsible for the highest number of verbal interjections. Once again, to avoid distortion of results, we examine the normalized data in Table 2. We observe that the Green Party has the highest number of verbal interjections per MP (similar to the case for nonverbal interjections). This aligns with Hitzler (1990),

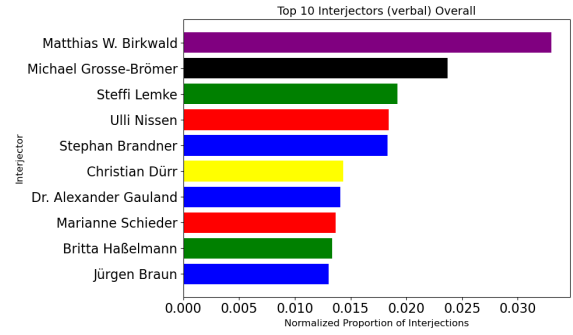


Figure 5: Top 10 Interjectors Overall

who found in 1990 that verbal interjections are particularly widespread within the Green Party, a trend that apparently persists to this day. Additionally, we observe a clear distinction between government and opposition parties, with the SPD and CDU/CSU occupying the lowest positions, apart from the non-affiliated MPs.

Party	Count	Num. of MPs	Interj. per MP
GRUENE	15017	67	224.13
AfD	16628	88	188.95
DIE LINKE	12680	69	183.77
FDP	11897	80	148.71
SPD	12537	152	82.48
CDU/CSU	13619	246	55.36
fraktionslos	6	7	0.86

Table 2: Verbal Interjections by Party (19th period)

Top Verbal Interjectors

Inspecting the top 10 verbal interjectors of the 19th legislature period (Fig. 5), we see that six of them were Parlamentarische Geschäftsführer (Parliamentary Managers) of their party at the time (Birkwald, Grosse-Brömer, Lemke, Schieder, Haßelmann, Braun), two were stellvertretende Fraktionsvorsitzende (Deputy Group Chairpersons) (Gauland, Dürr), and only two were neither (Brandner, Nissen). This aligns with what Vögele and Thoms (2019) found, that politicians in higher positions are particularly active in making verbal interjections. The question remains whether they aim to profile their party and themselves, or whether their proximity to the stenographer makes them easier to be identified.

4.2 Gender Analysis

Table 3 provides interesting insights into the behavior of male and female MPs. Overall, men deliver a higher number of paragraphs during de-

bates and both receive and give more interjections. When these counts are normalized by the number of MPs of the respective gender in parliament, men still dominate in most categories. The only exception is for nonverbal interjections received, where women have a higher number per MP. The normalized number of verbal interjections given is nearly identical for men and women, aligning with findings from [Irmer and Versloot \(2024\)](#) in Baden-Württemberg, and contrasting with results from the Austrian parliament ([Hofmann and Wissik, 2024](#)) and the US Congress ([Miller and Sutherland, 2023](#)), where men dominate as interjectors.

Furthermore, women receive fewer verbal interjections and more nonverbal interjections. Considering that most verbal interjections tend to be critical ([Burkhardt, 1993](#); [Vögele and Thoms, 2019](#)), while most nonverbal interjections signify applause, these findings appear to be consistent with [Ash et al. \(2024\)](#) in state parliaments: Women receive more positive and fewer negative reactions than men.

It is also worth noting that in 11.6% of verbal interjections, the interjector (and thus also the gender) is unknown. This uncertainty might contribute to more instability or inconsistency in the results.

In conclusion, based on the frequencies of interjections alone, we do not observe significant differences in how interjections are used by and directed against female MPs.

Metric	Men Total (Per MP)	Women Total (Per MP)
Seats in Parliament	486 (-)	223 (-)
Num. paragraphs	201,210 (414.01)	80,685 (362.62)
Verbal interj. received	62,970 (129.57)	19,603 (87.9)
Verbal interj. given	50,741 (104.4)	22,775 (102.13)
Nonverbal interj. received	125,988 (259.23)	60,305 (270.42)

Table 3: Gender Distribution in Parliamentary Speech and Interjections. Values in parentheses are normalized by the number of MPs for each gender.

4.3 Speechact Classifier

While the analysis of interjections provides valuable insights into interactions and interjections in parliament, understanding the underlying nature of these interactions is just as crucial. While we have explored the frequency and distribution of verbal and nonverbal interjections, there is still the need to identify what exactly is being expressed through these interjections.

While human annotators can identify verbal interjections, manually classifying the specific speech act—such as whether it is an expression of agreement, criticism, or a rhetorical question—for

whole sessions or even whole legislature periods would be too time-consuming and resource-intensive. Here, LLMs could be leveraged to facilitate this task by automatically classifying speech acts based on the context of the interjections. However, this approach is still relatively rare in research on parliamentary speech.

One example of such an approach is the use of a BERT-based model to detect speech acts in German parliamentary speeches ([Reinig et al., 2024](#)). For their classification task, they distinguished, at the lowest level, between 14 classes and report an average F1-score of around 82%. These 14 speechacts are classified as follows:

Cooperation

- Regulatory:
 - Macro (1), Expressive (2)
- Informative:
 - Report (3), Question (4) ³
- Consolidating:
 - Request (5), Support (6)
- Other:
 - Subjective Statement (7) ⁴

Conflict

- Declarative:
 - Self-Representation (8), Promise (9)
- Confrontational:
 - Rhetorical Question (10), Accusation (11)
- Argumentative-Critical:
 - Demand (12), Rejection (13), Bad Outcome (14)
- Other:
 - Subjective Statement

Although their work focuses on classifying speech acts in utterances from parliamentary speeches, not interjections, I wondered whether it would be possible to apply this model to interjections as well. Additionally, I thought this model

³Note that for the model itself, [Reinig et al. \(2024\)](#) did not differentiate between informative and rhetorical questions.

⁴This class is denoted as 'evaluation' in the model.

could potentially be applied to speeches to gain insights into which speech acts might elicit interjections. For this reason, I used the BERT model with context fine-tuned by [Reinig et al. \(2024\)](#) to obtain predictions for speech acts in the debates. Given that preprocessing required tokenizing and preparing the speeches, which was time-consuming, only the last five debates of the 19th election period were chosen. It is important to note that while [Reinig et al. \(2024\)](#) trained a separate model to extract utterances from speeches and make predictions, we used entire sentences as utterances, effectively skipping that extraction step and treating the full sentence as the unit of analysis.

4.3.1 Evaluation

Since we are specifically interested in interjections, after obtaining predictions, we divided all textual data into four categories: interjections, utterances not followed by any interjections, utterances followed by verbal interjections, and utterances followed by nonverbal interjections. The corresponding plots can be found in the Appendix A.2. For all four cases, the classifier predominantly predicted the 'evaluation' class, i.e., subjective statements that, according to [Reinig et al. \(2024\)](#), "cannot be unambiguously classified as either cooperation or conflict" (p. 8289). They note that this class makes up a large proportion of sentences in their data, likely causing the model to be biased toward predicting it. Nonetheless, some interesting observations can be made. While for interjections, 93% of sentences are predicted as 'evaluation,' this number is smaller for utterances followed by nonverbal interjections (83%), verbal interjections (78%), and, lastly, utterances not followed by interjections (70%) (see Figure 27). This may indicate that the model struggles more to predict speech acts for interjections than for speeches, which makes sense given that it was not trained on interjections.

For all four cases, the second most predicted class is no prediction at all. However, when we only consider the remaining classes, subtle differences emerge between our four cases. For utterances followed by no interjections or verbal interjections, the next most predicted class is 'report,' while for utterances followed by nonverbal interjections, the next most predicted class is 'expressive.' This is interesting because 'expressive' generally refers to expressions of gratitude, which are often followed by applause—nonverbal interjections. However, it is challenging to draw more general conclusions

without considering gold labels.

Manual Annotations

To obtain closer insights, we manually annotated a sample of utterances. We quickly realized that applying the speech acts from [Reinig et al. \(2024\)](#) to interjections often did not make much sense, so we decided to annotate and evaluate only utterances from speeches. For this, we manually annotated 312 utterances with the speech act label and a Boolean indicator to note whether the sentence represented an utterance, as we noticed that some utterances were cut off due to erroneous tokenization or because the speaker was interrupted by interjections, so we wondered whether this incorrect utterance separation might have contributed to the high amount of 'evaluation' labels as well. We annotated 100 samples predicted by the model as 'evaluation,' 100 samples predicted as nothing, and at most 10 samples for each remaining predicted class, totaling 112 samples.

Utterances predicted as 'evaluation' While 40% of the utterances predicted as 'evaluation' were also labeled as 'evaluation' in our manual annotation, the model missed a lot of other classes, mostly 'report' (20%), 'accusation' (7%), 'expressive' (5%), 'question-all' (5%), and other classes that appeared in less than 5% of cases (see Figure 29). This shows that the model struggles to distinguish between a mere report and a subjective statement or a more hidden accusation, as parliamentary speech can be quite subtle. In approximately 84.4% of sentences, the sentence was also an utterance, while in 7.3% it was not, and in 8.3% the sentence was interrupted.

Utterances predicted as nothing For the utterances predicted by the model as nothing (see Figure 30), we only annotated the utterance as nothing in approximately 7% of cases where the sentence was so interrupted that no class could be attributed. In 25% of cases, the utterance was actually expressive (e.g., 'Thank you'), while in 20% it was a report, 11% a demand, and 6% a promise. Other classes were present in less than 5% of cases. It is puzzling that the model predicted many expressive utterances as nothing, especially since these are simply expressions of thanks. Similarly, the model struggles to detect questions (as seen in utterances predicted as 'evaluation'), even though these should be easy to identify, especially since they are marked with a question mark. In these cases, the sentence was an actual utterance in 80%

of cases, 12% were interrupted, and 7% were not utterances.

Utterances predicted as remaining classes In cases where the model predicted a class other than 'evaluation' or 'nothing,' recall and precision were high (over 82%) for classes such as 'demand,' 'evaluation & report,' 'expressive,' 'report,' 'promise,' 'rejection,' 'request,' 'self-representation,' and 'support.' For 'accusation,' the model had high precision (90%) but low recall (45%), while for 'question-all,' it showed lower precision (60%) and higher recall (86%). An interesting pattern emerged where the model often predicted 'question-all' for utterances containing the word 'Frage' (question) or 'fragen' (to question) that appeared shortly before the actual question. Here are two examples:

"Weil wir gerade bei Steuern und Gerechtigkeit sind: Wer ausgerechnet den Kampf gegen Steuerbetrug lieber weiter mit Fax und Brief führen will, den **frage** ich wirklich ernsthaft:"

"Die **Frage** lautet:"

The same phenomenon occurred with 'promise':

"Die Bundesregierung gab den Hinterbliebenen der zwölf Getöteten und den vielen Verletzten ein **Versprechen**: die Aufklärungsarbeit des Ausschusses uneingeschränkt zu unterstützen."

It performed particularly poorly in terms of precision in 'macro' (14%) (but a recall of 100%) and in predicting the combined label 'accusation & bad-outcome' (0% recall, 0% precision). This subset focuses only on cases where the predictions are not 'evaluation' or 'nothing,' which explains why precision and recall are low for these classes—it is because this subset contains cases where the model never predicted. While it did over-generate predictions for 'evaluation' and 'nothing,' as we have seen above, it still missed some cases where these classes should have been predicted. All results can be found in Table 4 in the Appendix.

This evaluation, assisted by manual annotation, has shown that the model seems to struggle with understanding the subtleties of parliamentary speech (e.g., when accusing other parties). It also does not always appear to rely on utterance given as a whole but rather focuses on individual words like

'promise' and 'question.' For other classes, however, it shows robust results, meaning there is high variability across classes, and thus, relying on an average F1 score may not provide enough insight into its performance.

Furthermore, my initial suspicion that incorrectly separated utterances might be responsible for the high number of 'evaluation' predictions could not be confirmed, as the variability between correctly and incorrectly separated utterances across classes and performances was too high. One potential source of error is the imbalanced dataset the BERT model was fine-tuned on.

Additionally, we found that a speech act classifier fine-tuned on speeches cannot be easily applied to interjections without issues. In the future, it would certainly be interesting to train a model specifically for interjections. It should also be noted that the classification of speech acts can be highly subjective and context-dependent. While [Reinig et al. \(2024\)](#) had a team of four annotators, I did this alone using their annotation guidelines, so my annotations likely differ from theirs in some respects.

4.4 Differences in lexicon between utterances followed by verbal/nonverbal interjections vs. followed by none

Inspired by the comparison of speech act classifier predictions on utterances followed by verbal vs. nonverbal interjections, I explored the lexical differences between them. For this analysis, I used the TF-IDF vectorizer from scikit-learn ([Pedregosa et al., 2011](#)) with German stopwords and manually added stopwords related to parliamentary procedures⁵. This approach allowed me to identify the top 50 most important words in sentences followed by nonverbal interjections, verbal interjections, and sentences without any interjections. This analysis is based on the last five debates of the 19th legislative period.

The resulting Venn diagram, which shows the overlap between these top 50 words, can be found in the Appendix (Figure 31). Probably the most interesting finding is that all parliamentary parties, except for the Left Party, appear in the Venn diagram, either in the top 50 words for sentences followed by verbal interjections (the Greens, FDP, CDU, CSU) or in the overlap of the top 50 words

⁵"Kollegen", "Kolleginnen", "Liebe", "Präsidentin", "Präsident", "Lieber", "Herren", "ja", "Damen", "geehrte", "geehrter"

for both nonverbal and verbal interjections (AfD, SPD). This suggests that mentions of political parties elicit reactions from other members of parliament.

For sentences followed only by nonverbal interjections, words like "herzlichen", "verantwortung", "endlich", and "tun" appeared. These words indicate expressions of gratitude or possibly calls to action, which could elicit a positive reaction from the listeners (applause). In contrast, sentences not followed by any interjection contained less controversial words, such as "euro", "jahre", and "prozent", suggesting that these were part of more neutral or factual reports, not eliciting (m)any interjections.

In the overlap between all three types, we also find the word 'Afghanistan.' This is likely due to the fact that these sessions took place between June and September 2021, a period when the withdrawal of troops from Afghanistan had just occurred and was highly controversial, eliciting reactions.

4.5 Party Mentions Analyses

Since we have just seen that the mentioning of parties appears to elicit interjections, this final experimental section examines the role party mentions play in parliamentary interaction. To this end, we used a list of party names and other possible references to extract mentions of parties in speeches. The basis for this analysis is, once again, the last five debates of the 19th legislative period. While most cases were straightforward, the most challenging was "Union," which is often used to refer to the CDU/CSU. Here, we excluded instances where "Union" was followed by "europäische," i.e., excluding mentions of the European Union. After processing, we manually reviewed all occurrences of "Union" to ensure there were no additional outliers. This review uncovered only two cases where I manually corrected the annotations:

"Damit unterstützt die deutsche Bundesregierung, damit unterstützen wir als Deutschland insgesamt Erdogans nationalislamistisches Regime, das völkerrechtswidrig in Syrien einmarschiert ist und zusammen mit Aserbaidshan gegen Armenien zu Felde zieht, um letztlich eine **pantürkische Union** als neuen Machtfaktor in der Welt zu schaffen."

"Wir wollen eine klimaneutrale **Union**!"

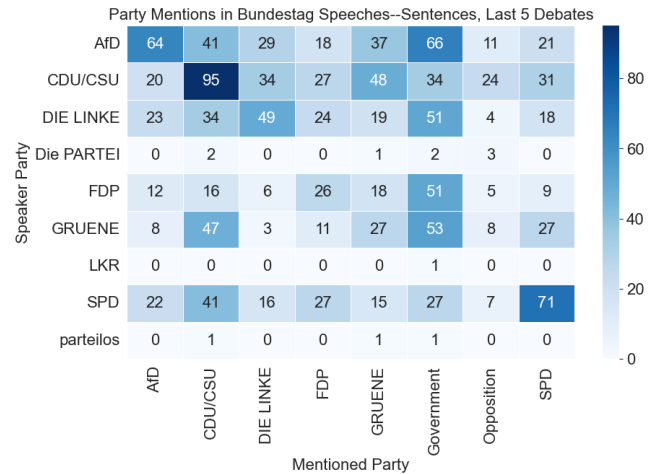


Figure 6: Heatmap: Num. of Sentences with Party Mentions

In the latter case, the speaker was referring to the European Union. The full list of terms used to detect party mentions is available in Table 5 in the Appendix. While this extraction process includes aliases and aims to capture as many party mentions as possible, the results may still contain false positives or fail to catch all mentions.

It was found that when speakers mention their own party, in 64% of cases, this does not elicit any immediate reaction. In 20% of cases, it is followed by a nonverbal interjection, in 10% by a verbal interjection, and in 5% by both types. By contrast, when another party is mentioned, 71% of sentences are not followed by any reaction. However, 12% elicit nonverbal interjections, 14% verbal interjections, and 3% both. These findings suggest that while mentions of a speaker's own party primarily elicit nonverbal reactions (e.g., applause) if any, mentions of other parties provoke slightly more verbal interjections (potentially as responses, counter-attacks, or defenses).

The number of sentences mentioning a party (or the opposition/government) is visualized in Figure 6. On the x-axis, the mentioned party is displayed, and on the y-axis, the party of the speaker mentioning it.

We can observe that opposition parties (AfD, the Left, FDP, and the Greens) predominantly mention the government in their sentences, whereas the government parties mostly reference their own groups. For most opposition parties (AfD, the Left, and FDP), their own groups are the second-most mentioned parties, whereas the Greens most frequently mention the CDU/CSU after the govern-

ment. Likewise, the CDU/CSU frequently mention the Greens, while the SPD primarily reference the CDU/CSU. These debates, taking place before the general election, might reflect an election campaign mode: the government may highlight its achievements, while opposition parties critique the government to position themselves. However, based on frequencies alone, we cannot definitively conclude whether mentions between certain parties (e.g., Greens and CDU/CSU) reflect mutual attacks or potential coalition-building, although the former seems more likely. Similar ambiguity exists for SPD mentions of their coalition partner—do these emphasize shared accomplishments or contain accusations?

4.5.1 Speech Acts and Interjection Types

For a more fine-grained analysis, we manually annotated sentences from the last five sessions in two categories:

- Sentences mentioning the speaker's own party and eliciting reactions.
- Sentences mentioning another party, with the mentioned party reacting.

These sentences were annotated using the speech act classes by [Reinig et al. \(2024\)](#) for the speeches themselves while potential verbal interjections were classified based on [Burkhardt \(1993\)](#), presented in Section 2. This yielded 144 annotated speech acts for sentences mentioning the speaker's own party with reactions, and 88 for those mentioning other parties, along with any interjections.

Own Party Mentioned

As shown in Figure 7, when a party reacts to the mention of its own party, the speaker often employs the speech act of self-representation, focusing on their own (or their party's) achievements and values. Some examples include:

"Europa ist für ihn und uns als CDU/CSU-Fraktion eine echte Herzensangelegenheit."

"Und dafür steht Die Linke."

"Das wollen wir von Bündnis 90/Die Grünen ändern."

This speech act is predominantly used across all parties. However, while the AfD does employ

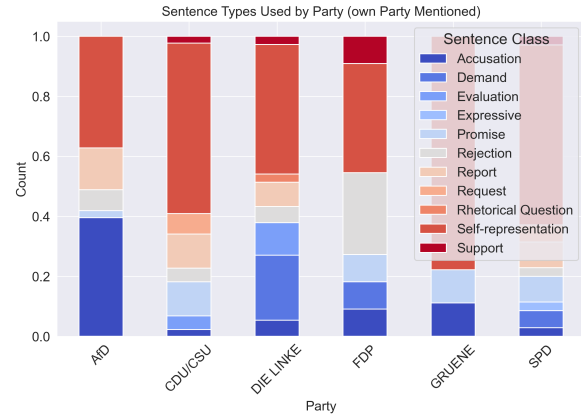


Figure 7: Speech acts for when own party is mentioned, and own party reacts

self-representation frequently, they often combine it with implicit or explicit accusations against other parties. For instance:

"Für diesen Satz steht in Deutschland 2021 nur noch die AfD."

"Die Alternative für Deutschland ist die einzige Partei im Parlament, die deutsche und nationale Interessen konstruktiv vertritt."

By using words like 'nur noch' and 'einzige' ('only'), the AfD not only seeks to enhance its own image but also to discredit other parties, accusing them of completely abandoning or betraying German and national interests. This supports the initial question of whether the AfD is isolated or deliberately isolates itself from other parties.

In contrast, the FDP frequently uses rejection to position itself as a strong opposition party:

"Das ist mit der FDP ganz eindeutig nicht zu machen."

When analyzing interjections received after a party mentions itself, two clear patterns emerge:

As shown in Figure 8, when a party mentions itself, the predominant reaction from its members is applause. For CDU/CSU and Die Linke, there are also verbal interjections from within the party, such as memoranda (1) and dissentiva (2):

1) *Bernhardt Loos [CDU/CSU, Speaker]:*
"er war nämlich von der Union –, forderte Wohlstand für alle –"

Stefan Müller [CDU/CSU, Interjector]:
"Der war auch kein Bayer, sondern ein Franke!"

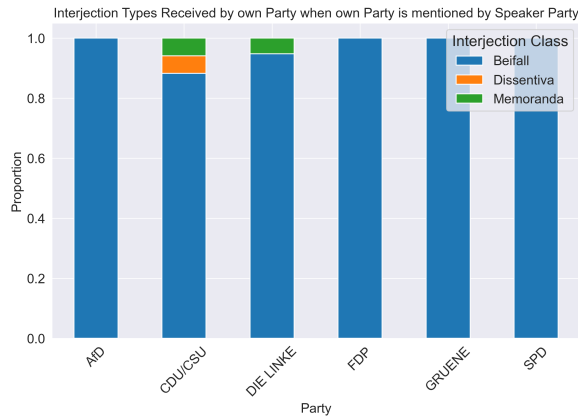


Figure 8: Interjection Types Received by Own Party when Own Party is Mentioned by Speaker Party

2) *Matthias Heider [CDU/CSU, Speaker]*: "Auch wenn die Union aus nachvollziehbaren Gründen dem Antrag heute nicht zustimmen kann – wir werden darüber in den nächsten drei Monaten noch mal nachdenken –"

Michael Grosse-Brömer [CDU/CSU, Interjector]: "Der reicht nicht aus!"

In the first example, Müller humorously highlights the rivalry between Bavarians and Francophones. In the second, Grosse-Brömer critiques the FDP's proposal while also disagreeing with his own party colleague, Heider, who suggests that the proposal should be reconsidered, as it is insufficient.

For the Left party, we observe a phenomenon that has not been addressed in this work so far: MPs often react to other interjections, not the speech itself (this is called 'Gegenruf' or counter-interjection). This can then develop into mini-dialogues as seen here:

Dieter Dehm [Left Party, Speaker]: "Liebe Grüne, lieber Jürgen Trittin, was würde Joseph Beuys zu den Auslandseinsätzen heute sagen?"

Peter Beyer [CDU/CSU, Interjector]: "Scheißegal, was der sagen würde!" –

Dr. Franziska Brantner [BÜNDNIS 90/DIE GRÜNEN, Interjector]: "Sind das die Nachwirkungen vom russischen Impfstoff, Diether?"

Dieter Dehm [Left Party, Speaker]: "Wir Linken sagen: Schluss damit!"

Peter Beyer [CDU/CSU, Interjector]: "War das ein Unsinn!" –

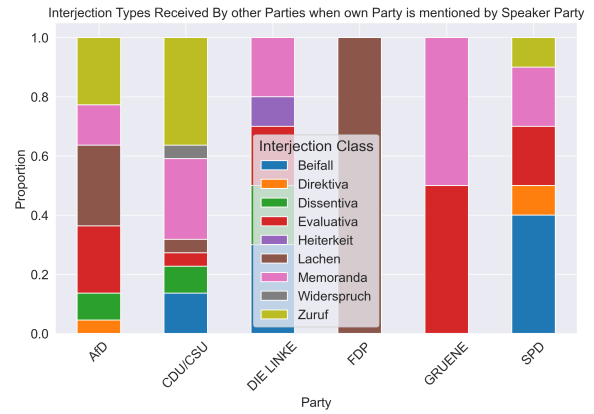


Figure 9: Interjection Types by Other Parties when Own Party is Mentioned by Speaker Party

Dr. Marie-Agnes Strack-Zimmermann [FDP, Interjector]: "Was hat denn Joseph Beuys damit zu tun? Der kann sich nicht wehren!" –

Dr. Diether Dehm [DIE LINKE, Interjector + Speaker]: "Der war bei den Grünen!" –

Dr. Franziska Brantner [BÜNDNIS 90/DIE GRÜNEN, Interjector]: "Sind das die Nebenwirkungen?"

Dehm interjects his own speech by participating in the mini-dialogue.

In contrast, reactions from other parties when the speaker mentions their own party display a broader range of interjections, as illustrated in Figure 9. Only the SPD, CDU/CSU, and the Left Party receive applause from other parties when mentioning themselves.

A detailed visual breakdown (see Figure 32 in the Appendix) shows, for example, that the SPD receives applause from their coalition partner CDU and ideologically close Greens. Meanwhile, they also face negative verbal interjections from the AfD, FDP, and their coalition partner. Interestingly, the CDU does not receive applause from the SPD, suggesting that smaller coalition partners may be more critical of their larger partners than vice versa, as noted by [Vögele and Thoms \(2019\)](#).

The AfD receives negative reactions across the board, such as laughter (ridicule) from all parties. On the other hand, the Greens face frequent verbal attacks from the AfD, accounting for half of the negative interjections they receive.

In the co-occurrence matrices (Figures 10

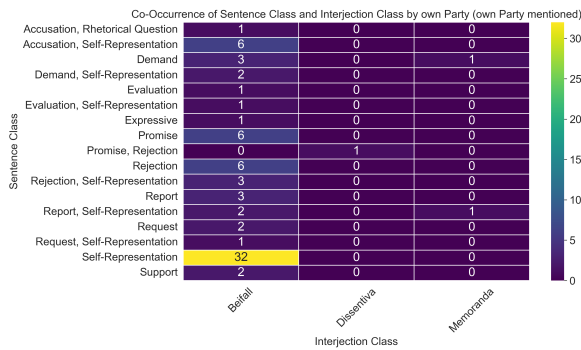


Figure 10: Co-Matrix of Interjection Types by Own Party when Own Party is Mentioned by Speaker Party

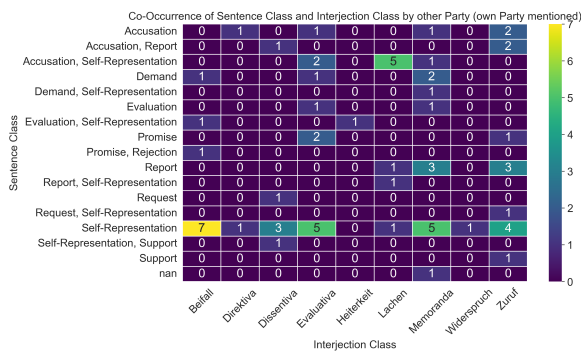


Figure 11: Co-Matrix of Interjection Types by Other Parties when Own Party is Mentioned by Speaker Party

and 11), we see a summary of these interactions: The most common co-occurrences are self-representation accompanied by applause from one's own party, while interjections from other parties show a more diverse distribution, including both support and critique.

Other Party Mentioned

Now we turn to sentences where another party is mentioned by the speaker party and reacts to it. As seen in Figure 12, when other parties are mentioned, it is mainly to accuse the other party.

This is most prominent for the Greens, SPD, and AfD.

"Die ökosozialistischen Grünen und auch Sie als Linke sind letztlich die Sterbebegleiter des deutschen Sozialstaats." [AfD]

"Rechtsextremismus und die Verbreitung von Hass über digitale Medien werden meines Erachtens auch von einer hier anwesenden Partei, nämlich der AfD und ihren Mitgliedern, unterstützt." [The Green Party]

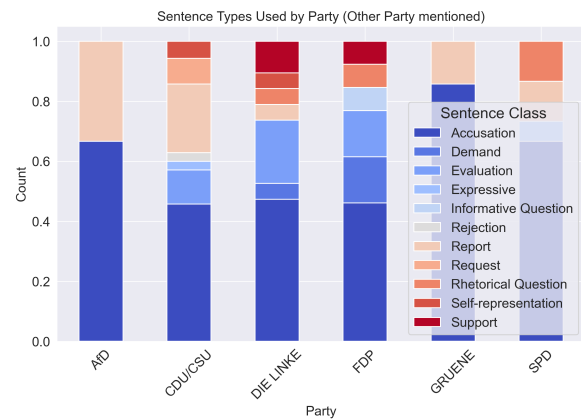


Figure 12: Co-Matrix of Interjection Types by other Parties when own Party is mentioned by Speaker Party

"Wenig überraschend hingegen ist, dass AfD, FDP und auch CDU/CSU mit manchen gemeinnützigen Organisationen und Lobbyorganisationen kein Problem haben, zum Beispiel „Stiftung Familienunternehmen“, die mit aggressiven Kampagnen die Interessen der Vermögenden vertreten." [SPD]

We see that the SPD does not hold back from accusing their coalition partner of having superior motives.

The only parties also expressing support, followed by positive reactions (applause) from the mentioned parties, are the Left and FDP (although with restrictions), as seen in the following examples:

"Auch von SPD und Grünen gibt es viele gute Ideen." [The Left Party]

"Einigen Punkten, die Grüne und Linke hier vorlegen, können wir Freien Demokraten sogar zustimmen." [FDP]

With the exception of these cases, we see that the reactions given by the mentioned parties are mostly negative (see Figure 13). All mentioned parties mainly react with memoranda, evaluativa, dissentiva, and interjections that were indistinct for the stenographer. The party reacting with the most "positive" reactions is SPD, with affirmativa and applause accounting for approximately 30%. This could indicate that the SPD is mentioned in a more positive way, but also that they react more often to positive mentions. For AfD, we see that they are the only party reacting with laughing when mentioned,

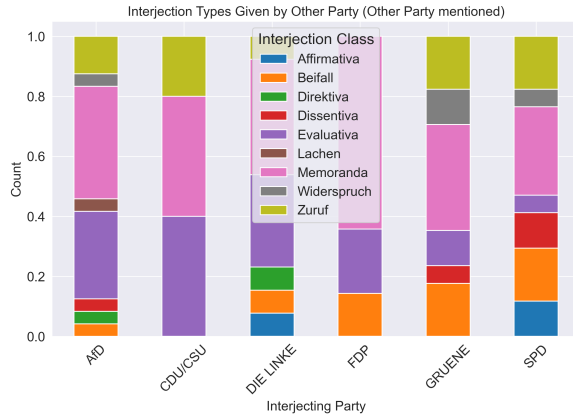


Figure 13: Interjection Types Given by mentioned Party to Speaker Party

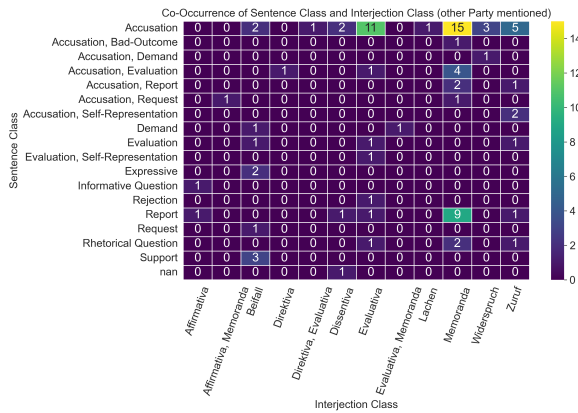


Figure 14: Co-Matrix of Interjection Types by other Parties when their Party is mentioned by Speaker Party

aligning with findings from [Vögele and Thoms \(2019\)](#), who found that MPs from AfD frequently use laughter to discredit political opponents.

Again, a more fine-grained visualization can be found in the Appendix, in Figure 33.

The co-occurrence matrix in Figure 14 presents an overview of all the data, showing that the reactions and speech acts are predominantly negative. This indicates that when another party is mentioned, it is often portrayed in a negative light, typically being accused of something. The mentioned party then responds, either to defend itself or to shift the blame back onto the speaker.

Particularly common patterns are accusation-evaluativa and accusation-memoranda.

1) *Bernhard Daldrup [SPD, Speaker]:*
"Das ist in Wirklichkeit ein Problem – das Thema Altschulden im Übrigen auch, zu deren Abbau Union und FDP leider im Moment nicht bereit sind."

Fritz Güntzler [CDU/CSU, Interjector]:
"Es gibt erfolgreiche Länder, beispielsweise da, wo die CDU regiert!"

2) *Detlef Seif [CDU/CSU, Speaker]:*
"Der AfD-Antrag macht an dieser Stelle mal wieder mehr als deutlich, dass es Ihnen letztlich egal ist, ob Menschen verfolgt sind oder nicht".

Unknown [AfD, Interjector]: "Das ist doch Blödsinn!"

In the first example, Güntzler uses memoranda to remind the speaker that federal states where the CDU is in government are successful in reducing debt. In the second example, the AfD interjector disqualifies the content of the speaker.

This analysis of party mention patterns revealed that self-mentioning is predominantly used to cast the speaker's own party in a positive light. Reactions from the speaker's own party typically come in the form of support (e.g., applause), though such reactions can occasionally be offered by other parties as well. However, negative responses, such as pejorative laughter or critical verbal interjections, often follow. In contrast, when speakers mention other parties and they react, the speaker primarily accuses the mentioned party of wrongdoing. Positive mentions of other parties are rare, and when they occur, the party being spoken of often responds with applause. In most instances, however, the mentioned parties are discussed negatively (accusations), and their reactions are similarly negative (evaluating or memoranda). Given that this sample is limited to the last five sessions of the legislative period, it would be valuable to compare this data to debates from earlier in the period. This could reveal whether the heightened confrontations are driven by the impending elections and whether coalition partners and/or opposition parties exhibit more support for each other outside the election campaign context.

5 Conclusions

In this work, we provided a broad overview of potential research areas in parliamentary interaction. We offer a ready-to-go tool to process XML files, as provided by [Blätte \(2017\)](#), along with inspirations and code for conducting analyses. These analyses are connected to related work, confirming previous studies on the same topic. Focusing on the 19th legislature period, we found that opposition

parties were more active verbal interjectors than governmental parties, with the Greens being particularly prominent (Hitzler, 1990). Moreover, we confirmed previous findings (Vögele and Thoms, 2019) that members of parliament holding high-ranking positions, such as parliamentary managers or group chairpersons, were among the top 10 interjectors.

When examining gender, we did not observe significant differences in how interjections are used by or directed against female versus male MPs, aligning with previous research (Ash et al., 2024; Irmer and Versloot, 2024). We tested a fine-tuned BERT model by Reinig et al. (2024), evaluating it with manual annotations. Our analysis revealed high variability across speech act classes and that the model cannot be easily applied to interjections. It would be interesting to train an interjection classifier based on the classes proposed by Burkhardt (1993).

Additionally, by enriching our analysis with figures and examples from the debates, we further explored party interactions. We showed that when parties mention themselves and there is a reaction, they tend to present themselves in a positive light, mostly receiving applause from their own side, with minimal support from other parties and more frequent criticism. In contrast, when other parties are mentioned and there is a reaction, they are often portrayed negatively, leading to counter-reactions where the speaker is reminded of unmentioned facts, corrected, or the content is simply disqualified.

Our focus on the AfD revealed that they are largely isolated, not receiving as much support from other parties as others do. However, they also isolate themselves by offering less support in return. Additionally, they set themselves apart by implicitly or explicitly attacking other parties, with the Greens appearing to be their primary target. Surprisingly, although the AfD were responsible for the highest number of verbal interjections, when normalized by the number of seats, the Greens emerged as the most active interjectors. Since this was only the first term for the AfD in the federal parliament and the right-wing populist party has since moved further right, becoming more extreme in certain areas, it would certainly be interesting to follow their trajectory.

Limitations and Future Work

There are still open questions and considerations that were previously disregarded. For instance, 'Zwischenfragen'—formal questions posed by MPs that require approval by the president—, as well as the corresponding answers have not yet been separated from the speeches. This is because they are encoded as part of the speech in the XML files provided by Blätte (2017), without any specific treatment. Similarly, mini-dialogues arising from back-and-forth exchanges between interjectors, which sometimes also involve the speaker, have not been accounted for separately. I suspect that these exchanges represent the majority of cases where MPs interject during speeches by their own party, although not exclusively, as we have already observed. Thus, an interesting next step would be to adjust the script to differentiate between speeches, formal questions, interjections, and mini-dialogues.

Additionally, it would be fascinating to explore which topics provoke larger numbers of interjections. As Vögele and Thoms (2019) found, controversial topics tend to elicit more verbal interjections. This analysis could likely be performed with the existing script, as it includes information on the agenda item that specifies the subject of parliamentary discussions.

While the script performs well (as verified through larger samples), there are still areas for improvement. For example, governmental actors are encoded differently from "regular" MPs; their titles appear without any mention of their party, often prefixed by terms like "Bundesministerin." Consequently, these actors are not currently captured by the regular expressions. Since only MP speeches (and not those by governmental actors) were used, this limitation was deemed acceptable for the current analysis. However, future work could include both speeches and interjections from governmental actors. By leveraging the Bundestag Stammdaten, it would be straightforward to associate their names with their respective parties.

Lastly, the XML files also indicate whether interjections came from some MPs within a party or the entire party (e.g., "Beifall bei Abgeordneten der + SPD" vs. "Beifall bei der SPD"). The script currently does not account for this nuanced information, as it has already addressed a sufficiently large number of dimensions for the scope of this paper. However, this distinction could provide valuable insights, particularly in detecting intra-party

disagreements on specific issues. This would be especially interesting in the context of the 20th legislative period, characterized by a three-party government and an early dissolution of the Bundestag. Future research could investigate whether certain parties, such as the SPD and Greens, were closer in alignment, whether the governing FDP behaved more like the opposition, and how patterns of disagreement and agreement evolved over time.

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A Appendix

A.1 Nonverbal Interjections

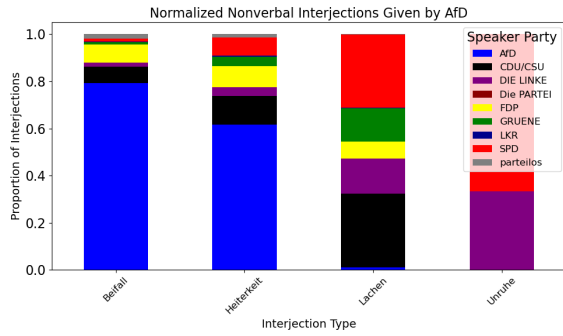


Figure 15: Nonverbal Interjections Given by AfD

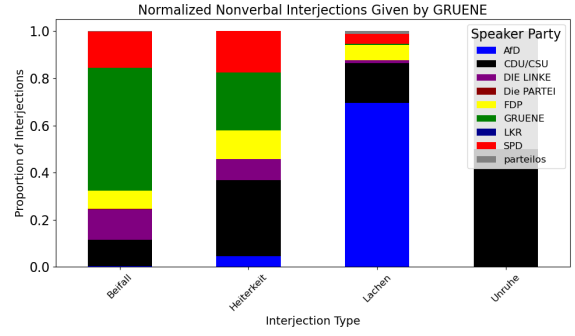


Figure 19: Nonverbal Interjections Given by the Green Party

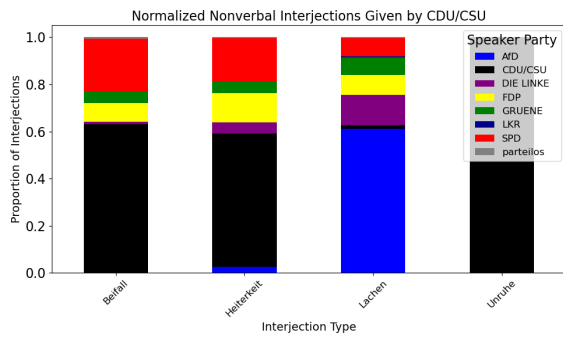


Figure 16: Nonverbal Interjections Given by CDU/CSU

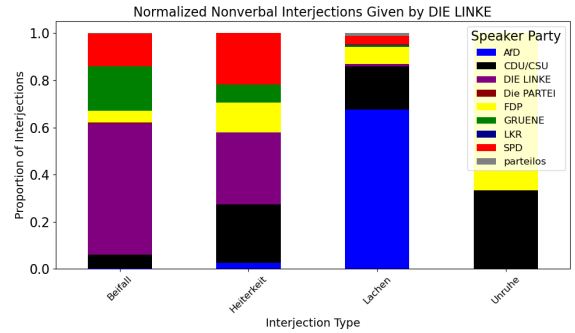


Figure 20: Nonverbal Interjections Given by the Left Party

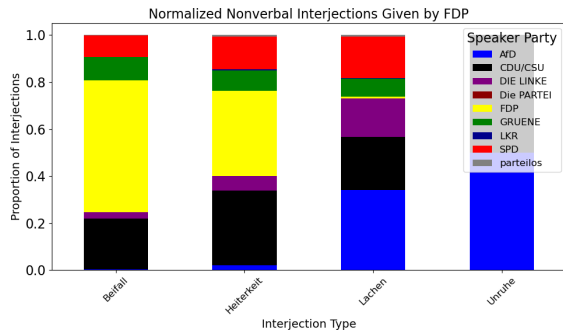


Figure 17: Nonverbal Interjections Given by FDP

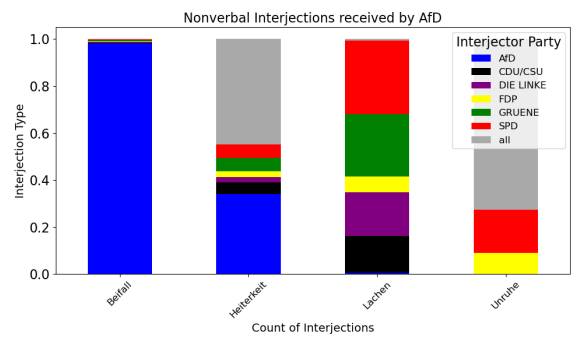


Figure 21: Nonverbal Interjections Received by AfD

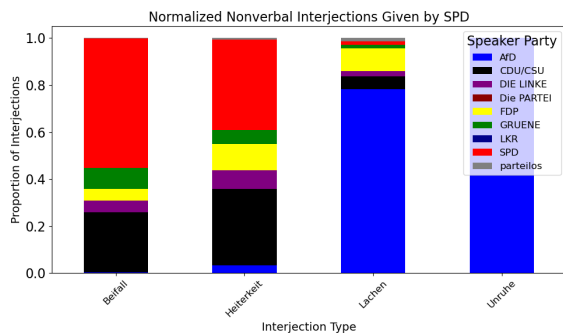


Figure 18: Nonverbal Interjections Given by SPD

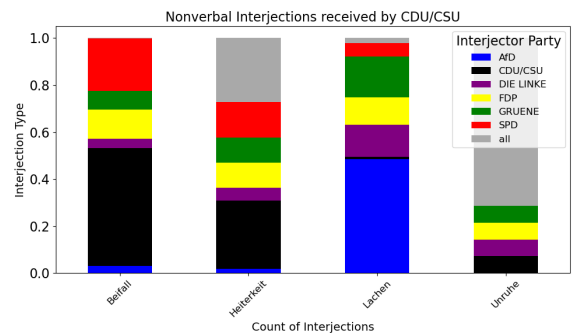


Figure 22: Nonverbal Interjections Received by CDU/CSU

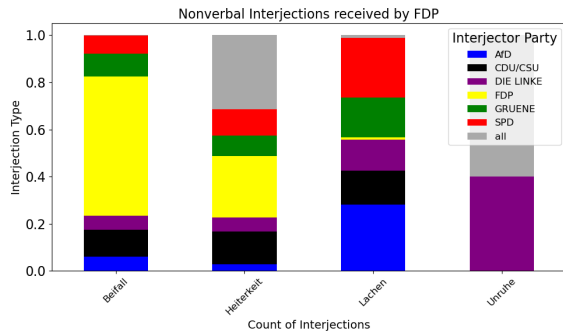


Figure 23: Nonverbal Interjections Received by FDP

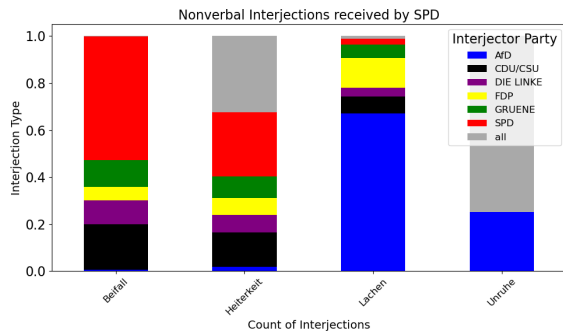


Figure 24: Nonverbal Interjections Received by SPD

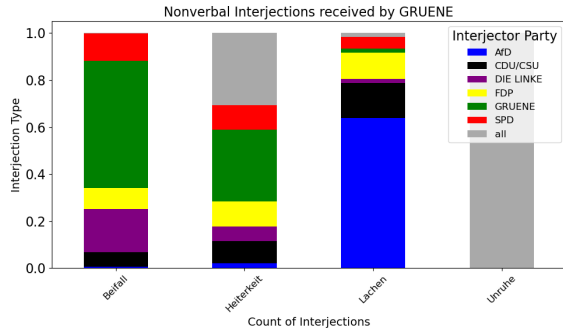


Figure 25: Nonverbal Interjections Received by the Green Party

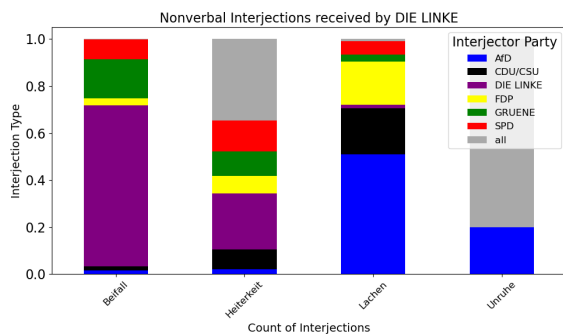


Figure 26: Nonverbal Interjections Received by the Left Party

A.2 Speechact Classifier Evaluation

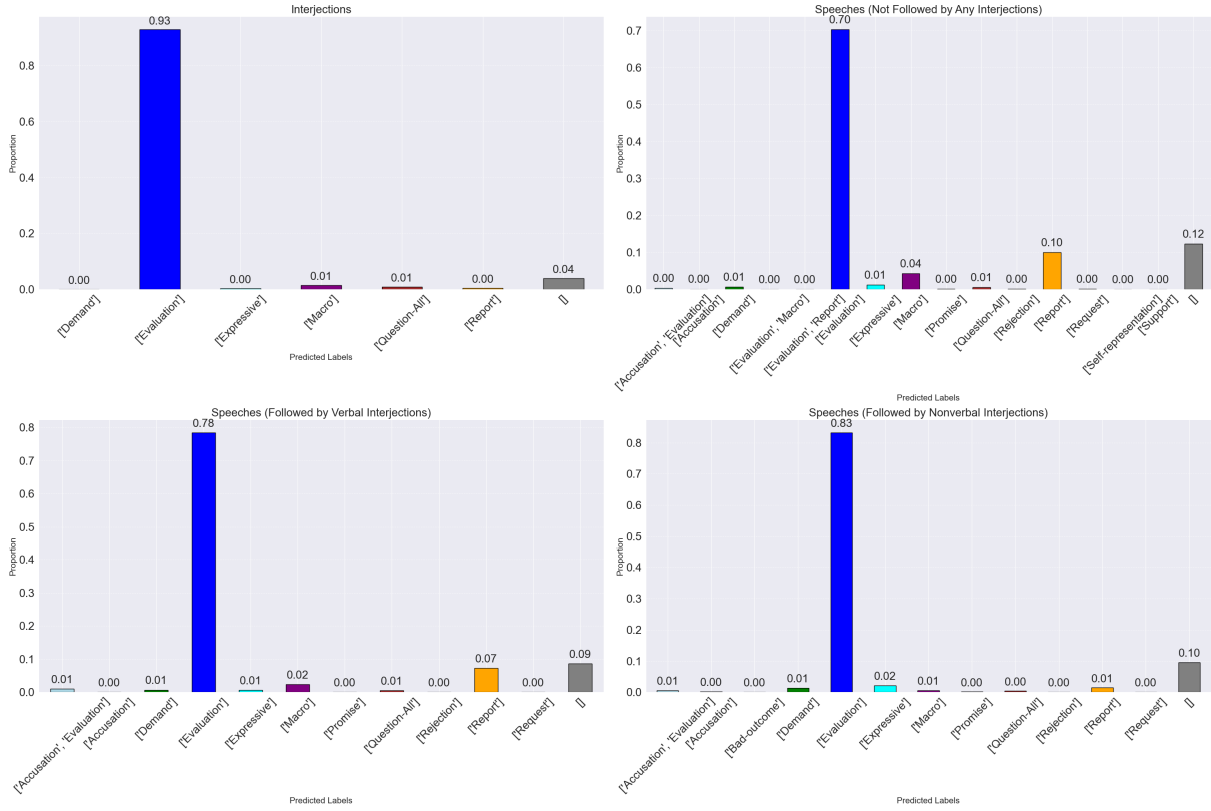


Figure 27: Predictions of classes obtained by BERT model (Reinig et al., 2024)

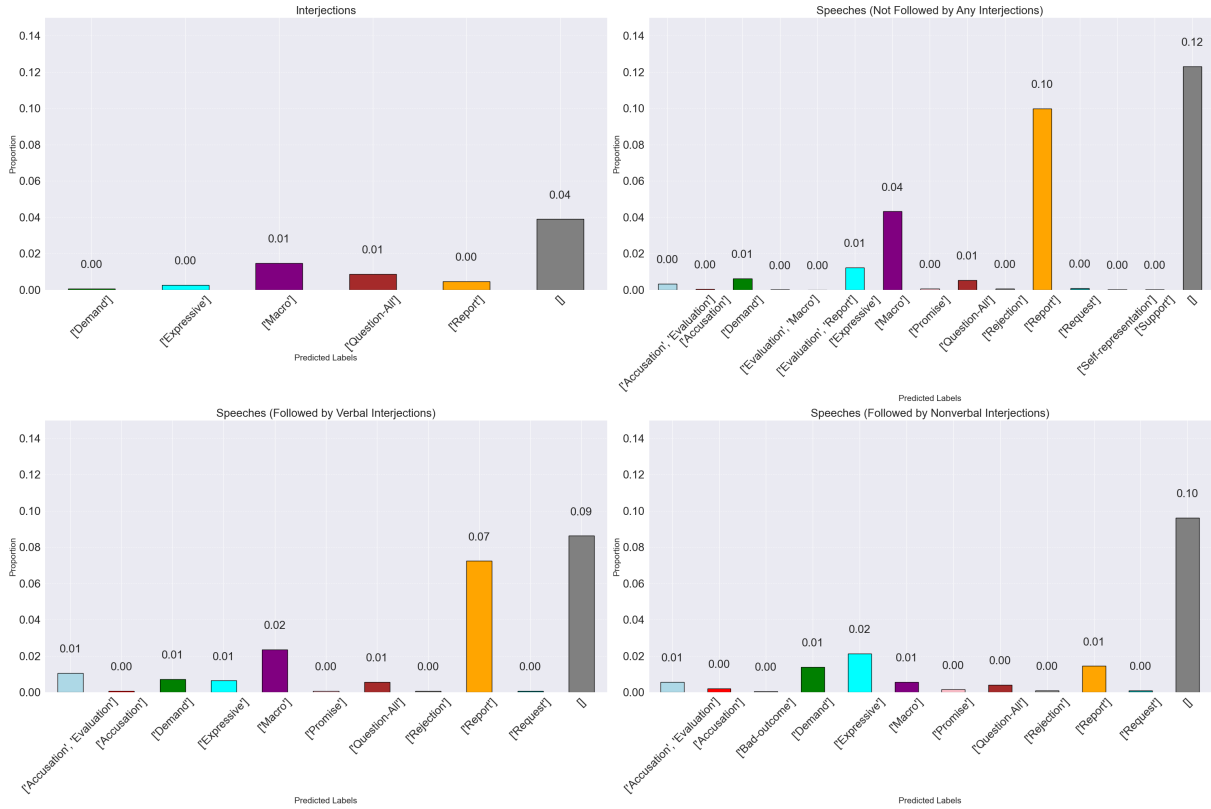


Figure 28: Predictions of classes obtained by BERT model (Reinig et al., 2024) (excluding 'evaluation')

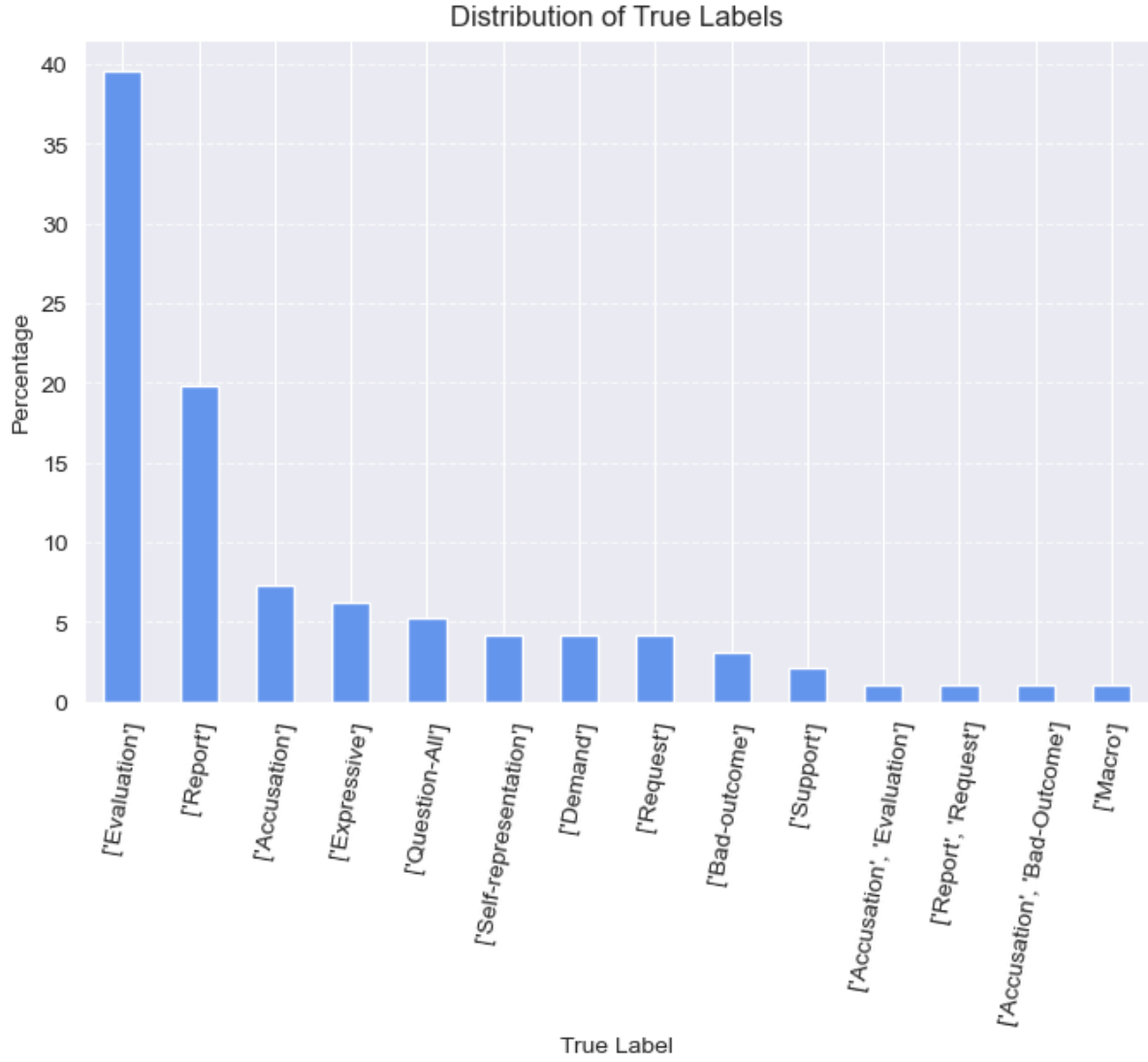


Figure 29: Gold labels for utterances the BERT model classified as 'evaluation'

Label	Precision	Recall	True (%)	False (%)	Interrupted (%)
Accusation	0.90	0.45	75.0	25.0	–
Nothing	0.00	0.00	–	–	100.0
Accusation, Bad-outcome	0.00	0.00	100.0	–	–
Demand	1.00	1.00	90.0	10.0	–
Evaluation	0.00	0.00	33.3	–	66.7
Evaluation, Report	1.00	1.00	–	100.0	–
Expressive	1.00	1.00	90.0	10.0	–
Macro	0.14	1.00	100.0	–	–
Question-All	0.60	0.86	100.0	–	–
Report	0.90	0.82	81.8	18.2	–
Promise	0.90	1.00	66.7	33.3	–
Rejection	0.90	1.00	100.0	–	–
Request	0.90	1.00	44.4	55.6	–
Self-representation	1.00	1.00	50.0	50.0	–
Support	1.00	1.00	100.0	–	–

Table 4: Performance Metrics and Utterance Distributions for Different Labels

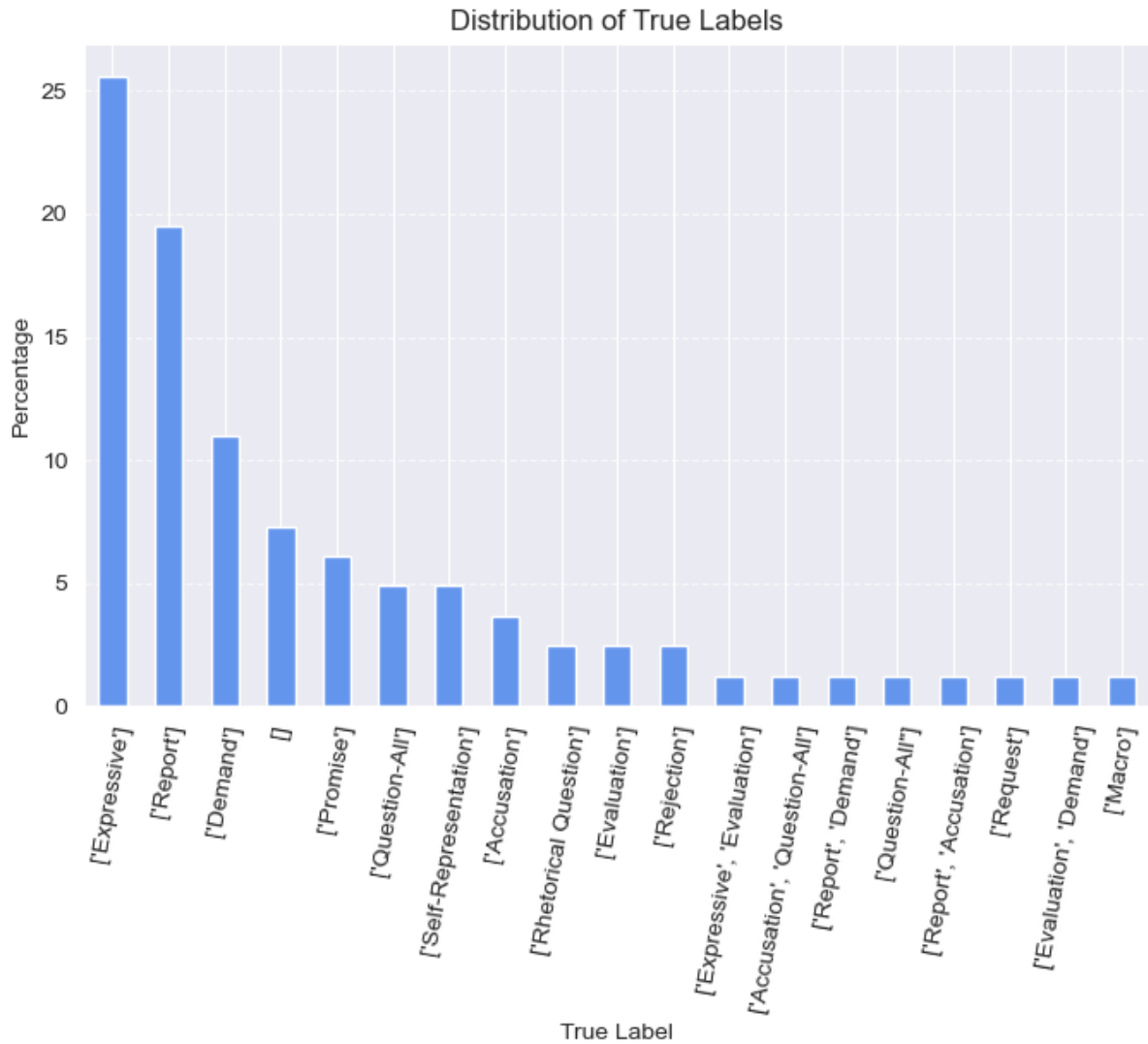


Figure 30: Gold labels for utterances the BERT model classified as nothing

Party	Aliases
AfD	AfD, Alternative für Deutschland
CDU/CSU	CDU, CSU, CDU/CSU, Unionspartei, Unionsparteien, Unionsfraktion, Unionsfraktionen, Union
FDP	FDP, Liberalen, Gelben, Gelbe, Liberale, Freie Demokratische Partei
SPD	SPD, Sozialdemokraten, Sozen, Sozis, Roten, Rote, Sozialdemokratische Partei Deutschlands
GRÜNE	Ökopartei, Grünen, Grüne, Bündnis 90/die Grünen
DIE LINKE	Linke, Linken, Linkspartei
Government	Groko, Regierungsparteien, Bundesregierung, Koalitionspartei, Koalitionsparteien, Koalitionsfraktion, Koalitionsfraktionen, Koalition
Opposition	Opposition, Oppositionspartei, Oppositionsparteien, Oppositionsfraktion, Oppositionsfraktionen

Table 5: Party Aliases

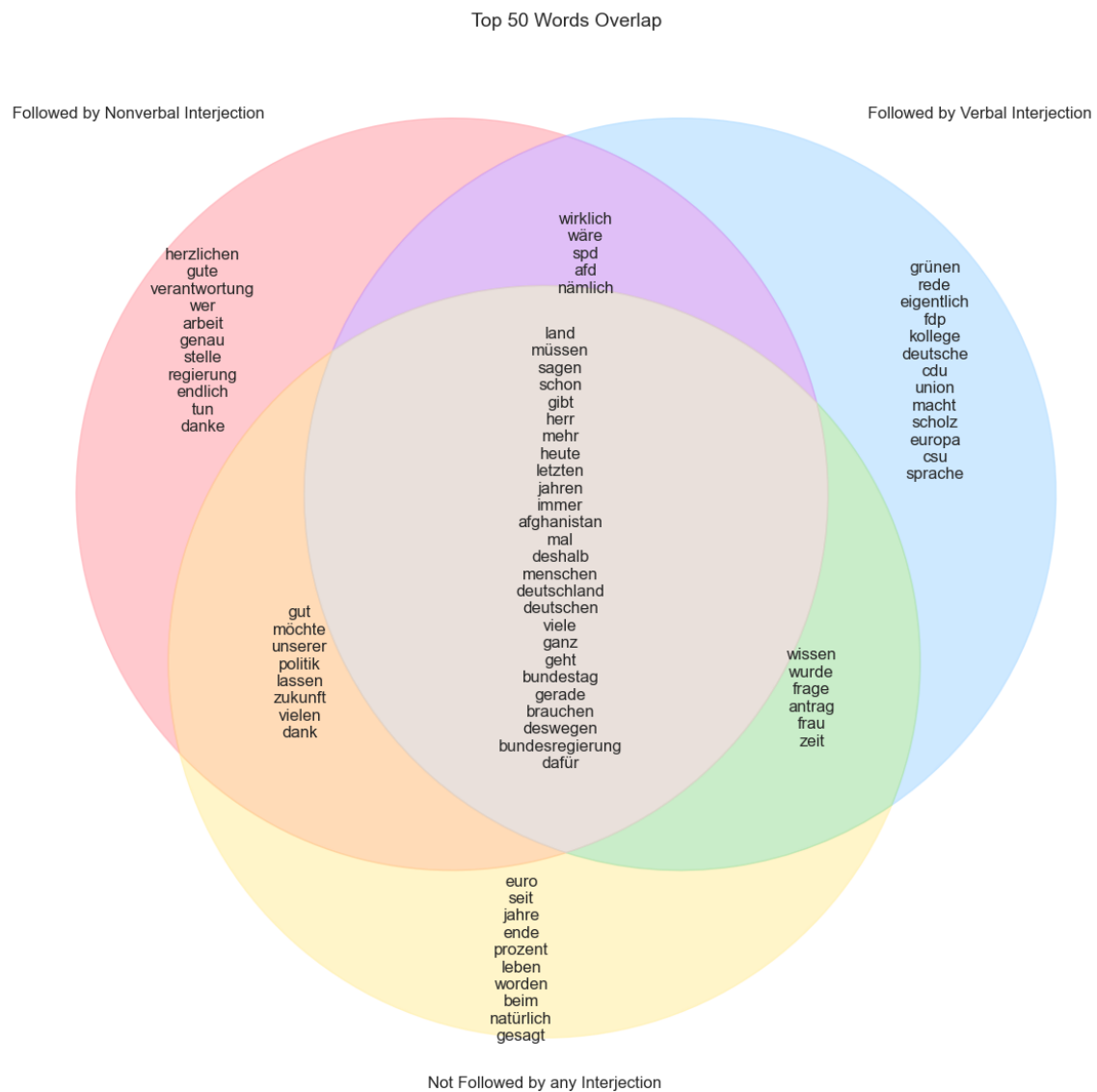


Figure 31: Overlap between Top 50 Words in Sentences followed by Nonverbal Interjections, by Verbal Interjections, by no Interjection

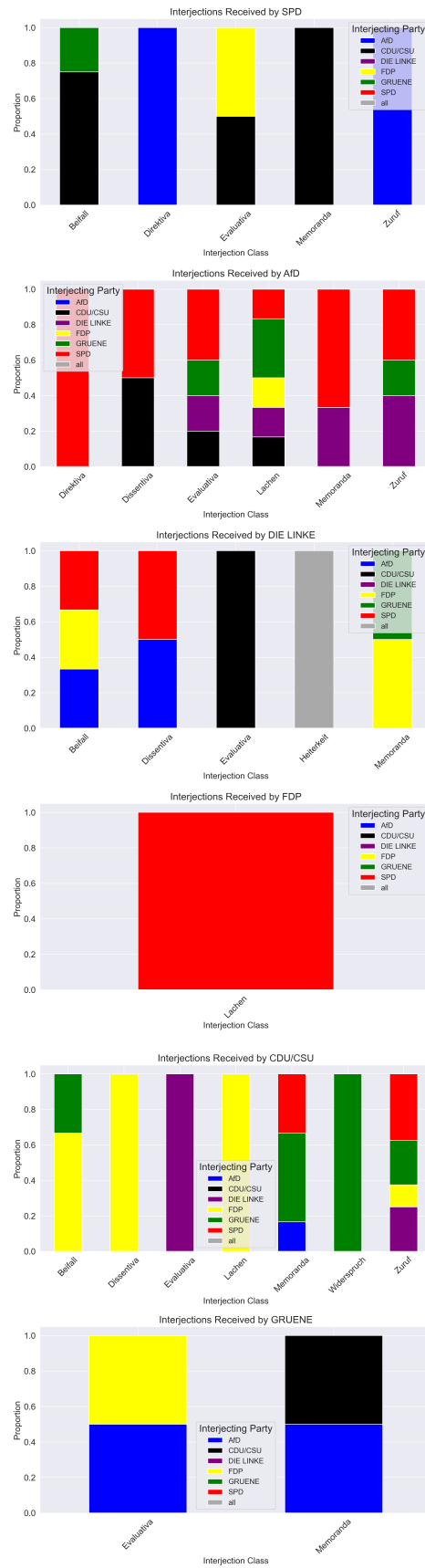


Figure 32: Fine-Grained Interjection Types Received when Own Party is Mentioned by Speaker Party

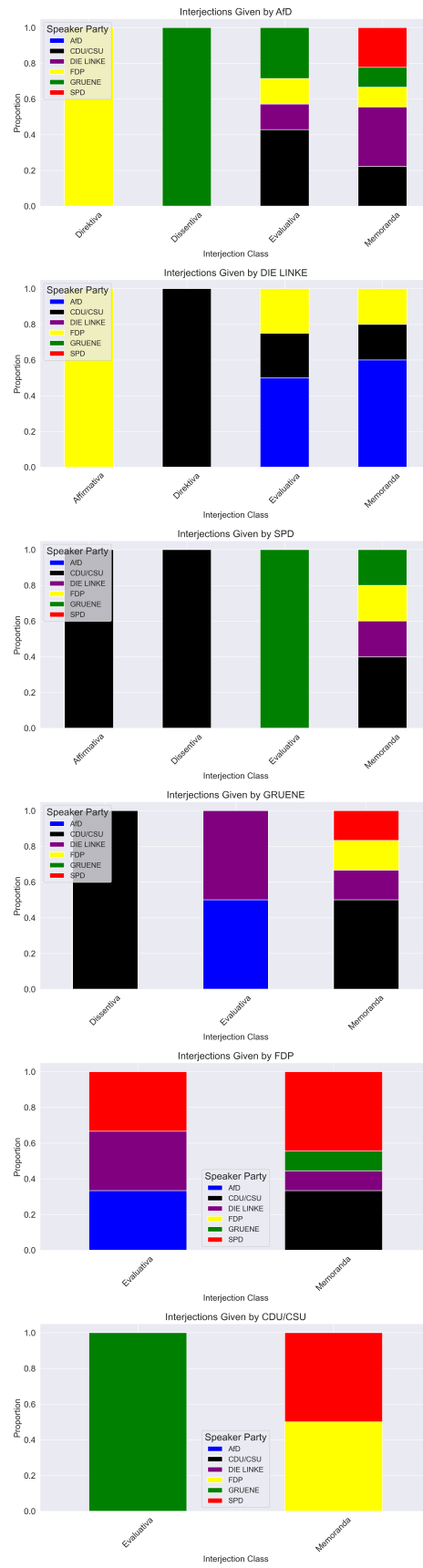


Figure 33: Fine-Grained Interjection Types Given when Another Party is Mentioned by Mentioned Party

A.3 Data extracted by the code

Column	Type	Description
Filename	String	Name of debate file
Period	String	Legislature Period
Date	String	Date in datetime format
Item	String	Name of agenda item of paragraph (e.g., <i>Antrag der Abgeordneten...</i>)
Speech #	Integer	Number of speech overall
Paragraph #	Integer	Number of paragraph within speech
Speaker	String	Name of the speaker
Role	String	Role of the speaker (e.g., MP, government)
Gender	String	Gender of the speaker (as noted in Bundestag Stammdaten)
Party	String	Party of the speaker
Paragraph	String	Content of the paragraph
Interjection	Boolean	True if the paragraph is an interjection
Interjector	String	Name of the interjecting person
Interjector Gender	String	Gender of the interjecting person
Interjector Party	String	Party of the interjecting person
Verbal interjection	Boolean	True if the interjection is verbal (e.g., <i>Zuruf</i> , <i>Widerspruch</i>)
Nonverbal interjection	Boolean	True if the interjection is nonverbal (e.g., <i>Beifall</i> , <i>Lachen</i>)
Interjection type	String	More specific interjection type (e.g., <i>Zuruf</i> , <i>Beifall</i> , <i>Lachen</i>)

Table 6: Description of the dataset columns